

1. Respiratory tract infections in adults (≥16y)

NICE 2023, NG237



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NICE guidance on triaging those ≥16y presenting with possible RESPIRATORY TRACT INFECTION (URTI/LRTI)

This page does NOT apply to those with COVID-19, nor to those <16y

REMOTE
ASSESSMENT



FACE-TO-FACE
ASSESSMENT



THINK: COULD THIS BE SEPSIS?
Assess as per NICE guidance on sepsis
and manage appropriately

NICE identifies the following RISK FACTORS for sepsis:

- Over 75y, or any age if frail.
- Recent surgery, trauma or invasive procedure (<6w).
- Diseases that impair immune system, including diabetes, asplenia, sickle cell.
- Drugs that impair the immune system: long-term oral steroids, other immunosuppressant drugs, cancer chemotherapy.

NICE sepsis guidance also highlights an increased risk of sepsis in those with skin breaches (e.g. wounds, catheter, iv drug use) and if pregnant/delivery in the past 6w, but, in these cases, sepsis is more likely from cellulitis/gynae symptoms than from respiratory infection.

Are symptoms/signs warranting further assessment present?
e.g. possible LRTI (new/increased breathlessness or confusion)

Arrange face-to-face assessment if:

- Cannot adequately assess remotely (e.g. communication difficulties).
- Serious illness suspected (e.g. pneumonia, non-infective causes).
- Comorbidity that may be exacerbated by a respiratory infection (e.g. frailty, COPD, immunosuppressed).

Urgency and place of assessment depends on severity/rate of deterioration.

If managing remotely, do NOT routinely prescribe antimicrobials unless the person knows when and how to seek further help AND there is sound reason.

Use clinical assessment to diagnose acute respiratory tract infection and decide if antimicrobials appropriate.

- If PNEUMONIA suspected: see section 3.
- Take comorbidity and social situation into account when making decisions about treatment/admission.
- Do NOT use point-of-care microbiological tests to decide if antimicrobials indicated.
- If flu-like illness, follow UKHSA advice on how to manage/antivirals.
- If unclear whether antibiotics needed, consider point-of-care CRP:

<20mg/L	Do not routinely give antibiotics
20–100mg/L	Delayed script?
>100mg/L	Offer antibiotics (see section 3)

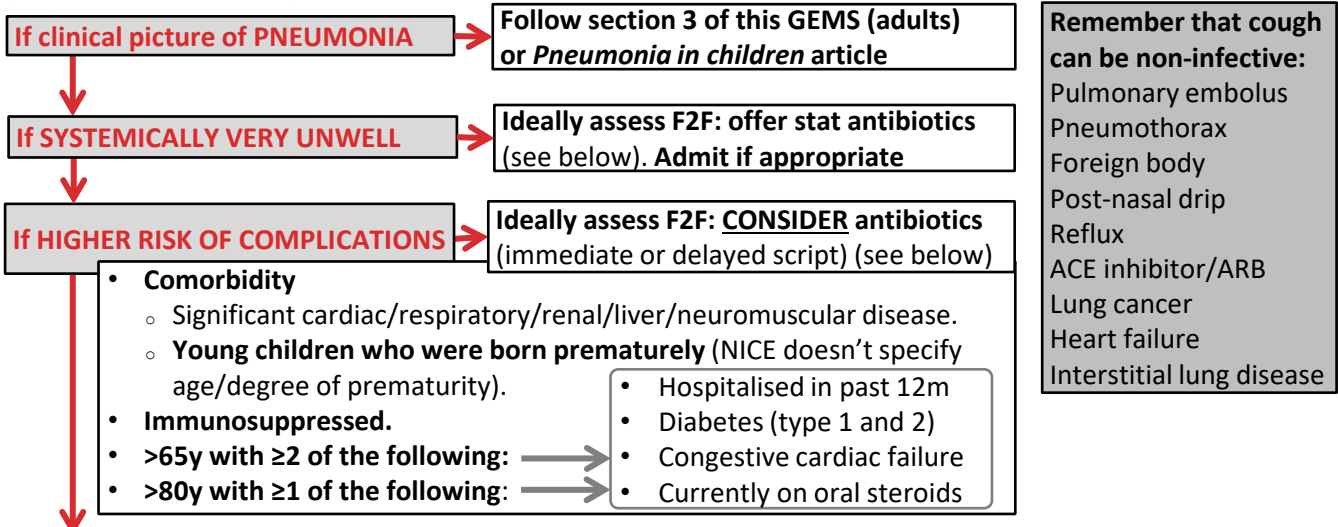
If this is something that can be managed at home with self-care:

Offer advice on self-management, including likely duration of illness and when/how to seek help (e.g. rapid deterioration, not improving over time).

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2. Acute cough in children and adults: causes, assessment and management (excluding COVID) (NICE 2019, NG120 and NICE 2023, NG237)

NICE reminds us of the infective causes of cough: pneumonia, acute bronchitis, URTI, viral-induced wheeze/bronchiolitis, croup, whooping cough (cough, cough, cough, cough, vomit!), acute exacerbation (infective/non-infective) of asthma/COPD/bronchiectasis. And remember non-infective causes!



NONE of the above: use clinical assessment to diagnose cause and manage appropriately. Remember that, especially in children, the causes include viral-induced wheeze, bronchiolitis, croup and whooping cough.

- Take comorbidity and social situation into account when deciding on treatment/if referral needed.
- Do NOT use point-of-care microbiological tests to decide if antimicrobials are indicated.
- If flu-like illness, follow UKHSA advice on how to manage/antivirals.
- If unclear if antibiotics needed, consider point-of-care CRP. If point-of-care CRP:

<20mg/L: do not routinely give antibiotics	20–100mg/L: delayed script?	>100mg/L: offer antibiotics
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(NICE doesn't refer to lab CRP – presumably because there is often a time lag before getting result)
- **If acute cough WITH URTI or ACUTE BRONCHITIS:** advise on self-care (see below). Do NOT offer antibiotics (even in acute bronchitis, they don't improve wellbeing and only shorten cough by half a day). Do NOT offer mucolytic, bronchodilators or inhaled/oral steroids (unless other indication, e.g. asthma).

Self-care (based on limited evidence, focusing on symptomatic relief, not shortening duration)

- **If >1y: honey** (trials hugely variable in terms of dose and frequency!).
- **If ≥12y:**
 - **Pelargonium** (herbal remedy; NICE doesn't specify dose) (weak evidence that symptoms/cough more likely to be better on day 7 compared with placebo: P.Sidoides used in trials).
 - **OTC expectorants containing guaifenesin** (in many OTC cough mixtures; pharmacy can advise).
 - **OTC cough suppressants** (but not those containing codeine).
- **No evidence for:** antihistamines, decongestants, codeine-containing preparations.

Expect cough to last 3–4w. If symptoms deteriorate significantly or rapidly, reassess and consider:

- Alternative diagnoses/signs of more serious illness (pneumonia, sepsis, cardiorespiratory failure).
- Previous antibiotic use that may have led to resistant organisms.

Antibiotics (if indicated) Note: it is '1 st line/alternative 1 st line', giving more scope to use the alternative than if '1 st /2 nd line'				
ADULTS		All courses are for 5d		
1 st line	Doxycycline	200mg on day 1, then 100mg once daily for 4d (<i>not if pregnant/breastfeeding</i>)		
Alternative 1 st line	Amoxicillin	500mg 3x daily (<i>preferred in pregnancy</i>)		
	Clarithromycin	250–500mg 2x daily		
	Erythromycin	250–500mg 4x daily (<i>preferred in pregnancy if macrolide needed</i>)		
CHILDREN		All courses are for 5d		
1 st line	Amoxicillin	1–11 months	1–4 years	5–17 years
		125mg 3x daily	250mg 3x a day	500mg 3x day
Alternative	NICE suggests clarithromycin as an alternative (or doxycycline if ≥12y). Doses as per BNF			
First-line choice: why does NICE recommend doxycycline for acute cough in adults vs. amoxicillin for pneumonia? For cough in adults, there is <u>limited</u> , <u>indirect evidence</u> that doxycycline is <i>slightly</i> better than other options (overall, antibiotics got people better 0.6d sooner). More importantly, NICE wants amoxicillin reserved for more serious infections to minimise resistance to amoxicillin.				

3. Pneumonia (adults only, not COVID related) (NICE 2019 NG120, NG138) (NICE 2023 NG237, CG191)



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NICE definitions for lower respiratory tract infections

NICE defines PNEUMONIA as: symptoms and signs of a lower respiratory tract infection (LRTI) and the GP (without a CXR) thinks it is pneumonia (because of focal chest signs, illness severity or other features).

➔ For children, see our article *Pneumonia in children* ➔

Clinical diagnosis of community-acquired pneumonia in ADULTS (without sepsis)

Use clinical judgement and CRB65 score to assess mortality risk and guide management

Score 1 for each CRB65 criteria:

Confusion: abbreviated mental test score $\leq 8/10$ or new disorientation in person/place/time

Respiratory rate: ≥ 30 /minute

BP: SBP < 90 mmHg or DBP ≤ 60

65: ≥ 65 y of age

CRB65 0: $< 1\%$ mortality risk
CRB65 1 or 2: $1-10\%$ mortality
CRB65 3 or 4: $> 10\%$ mortality

CRB65 = 0 (mild)

Most can be managed at HOME

Investigations

- Do NOT do sputum culture (CXR discussed only in context of admission)

Antibiotics

- 5d course of single antibiotic (see below)

Safety-net: ensure patients seek advice if:

- Symptoms worsening
- Symptoms not improving after 3d

CRB65 = 1

Shared decision-making about place of care: e.g. supported home care, virtual ward, hospital-at-home team

CRB65 ≥ 2 (moderate-severe)

Consider ADMISSION

For those admitted:

Investigations: CXR, blood & sputum culture, consider pneumococcal and legionella urinary antigen tests

Antibiotics

- 5d courses for most
- Consider dual therapy (amoxicillin+macrolide)
- Co-amoxiclav or cephalosporin may be used in more severe infection

Antibiotics for pneumonia (CRB65 0 or 1)

- Antibiotics as per table below:
 - All courses are for 5d
 - If clinically unstable on day 5 (e.g. fever in past 48h/SBP < 90 /P > 100 /RR > 24 /sats $< 90\%$ on air), NICE says continue antibiotics – and we think consider admission!
- Do NOT use steroids unless there is another indication (e.g. asthma, COPD)

ADULTS

All courses are for 5d

1 st line	Amoxicillin	500mg 3x daily (preferred in pregnancy) If CRB1 + atypical organism suspected, co-prescribe clarithromycin/erythromycin
Alternative 1 st line	Doxycycline	200mg on day 1 then 100mg once daily for 4d (not if pregnant/breastfeeding)
options	Clarithromycin	500mg 2x daily
	Erythromycin	500mg 4x daily (preferred in pregnancy)

Note: it is '1st line/alternative 1st line', giving more scope to use the alternative than if it was '1st/2nd line'

Rough guide for patients on recovery after pneumonia (also affected by severity/age/comorbidity):

4w: chest pain and sputum significantly reduced

6w: cough and breathlessness significantly reduced

3m: most symptoms have resolved, may be fatigued

6m: most people feel back to normal

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